Abbas Rizvi

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Education

University of Alberta — Expected 2028
BSc Computer Science | GPA: 3.67 — *Edmonton, AB*

Experience

Machine Learning Developer — Dec 2025 – Present
ChromacareLabs — *Edmonton, AB*
Skills: *Python, PyTorch, YOLO-OBB, ONNX, FastAPI, MLflow, AWS (SageMaker, ECS, RDS, S3), Docker, PostgreSQL*

- Architected a cloud deep-learning regression pipeline that quantifies a biomarker from at-home test-kit photos: rotated-box cassette detection, 1D intensity-profile extraction, and neural-network regression, deployed on AWS.
- Engineered models that generalize to unseen test units and production batches via custom architectures, calibration logic, and cross-batch validation, plus an ensemble that outputs calibrated prediction intervals.
- Built HDF5 / PostgreSQL (RDS) / S3 data infrastructure, API endpoints, and visualization tooling to track prediction consistency and data quality; automated model deployment from the MLflow registry to SageMaker.
- Work with chemists to feed signal-stability, failure-mode, and lot-to-lot analysis back into assay and cassette design; support FDA and Health Canada submissions with validation and reproducibility documentation.

ML Research Assistant — May 2025 – Dec 2025
Servier Virtual Cardiac Centre | University of Alberta — *Edmonton, AB*
Skills: *Python, PyTorch, NumPy, Pandas, Matplotlib, Git, HPC*

- **First author** of an IEEE BIBE 2025 paper benchmarking nnUNet, pretrained models, and a proposed transfer-learning approach for right-ventricular MRI segmentation; our approach achieved the best results.
- Building regression models (linear regression, gradient boosting) on clinician-reviewed data to estimate right-ventricular volume from long-axis measurements and fewer short-axis slices than full integration.

Teaching Assistant — Sep 2025 – Dec 2025
CMPUT 201: Practical Programming Methodology | University of Alberta — *Edmonton, AB*
Skills: *C, Git, Linux*

- Led weekly C-programming labs, developed assignments and in-class exercises with sample solutions, and graded assignments and exams with detailed feedback.

Data Analysis Assistant — Feb 2025 – Aug 2025
Faculty of Medicine & Dentistry | University of Alberta — *Edmonton, AB*
Skills: *SPSS, Excel*

- Analyzed survey data across medical specialties to find patterns in pain management for dementia and where providers agree or disagree; **co-authored** a research article and a poster presented at the Canadian Conference on Dementia.

Projects

3D DICOM Cardiac MRI Visualizer | GitHub
Skills: *C++, VTK, Qt*

- Built a tool that reconstructs 3D DICOM volumes from 2D slices using position, spacing, and orientation metadata, and maps anatomical contours onto the volume.
- Added interactive viewing of DICOM time series so users can select slices and inspect volumetric changes.

Cardiac LGE-MRI Segmentation | GitHub — Sep 2026 – Present
Skills: *Python, PyTorch, nnU-Net, SLURM*

- Benchmarking nnU-Net v2 variants on the EMIDEC cardiac LGE-MRI dataset with stratified 5-fold cross-validation, segmenting heart scar tissue under extreme class imbalance (scar is under 0.2% of voxels).
- Developing a second-stage U-Net from scratch that re-segments the heart in polar coordinates, using circular convolutions and seam-aware upsampling to refine nnU-Net predictions.
- Built a reproducible evaluation suite with size-based post-processing gates chosen by threshold sweeps; run training as SLURM job arrays on a GPU cluster.

City of Edmonton Property Value Predictor | GitHub
Skills: *Python, XGBoost, scikit-learn, Pandas, Flask*

- Integrated open-data and OpenStreetMap features for 360,000 Edmonton residences; 5-fold XGBoost model reached R^2 = 0.917 and MAE = $21.5K, deployed as a Flask app on Render.

Publications

Transfer Learning for Right Ventricular MRI Segmentation. A. Rizvi (first author) et al. IEEE BIBE 2025. PDF

Technical Skills

Languages: Python, C, C++
ML & Data: PyTorch, XGBoost, scikit-learn, OpenCV, ONNX, MLflow, NumPy, Pandas, Matplotlib, SPSS
Cloud & Tools: AWS (SageMaker, ECS, ECR, RDS, S3), Docker, FastAPI, Flask, PostgreSQL, Git, Linux, HPC